Purple-Team Detection Lab

Measuring Zeek's Network Detection Coverage Against MITRE ATT&CK

Asmi Mishra · October 2026

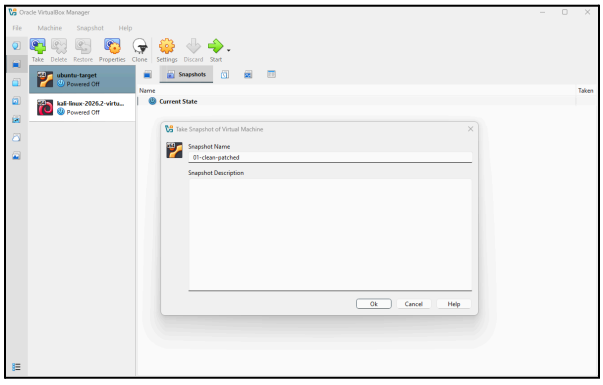
GitHub: github.com/AsmiMishra/purple-team-detection-lab

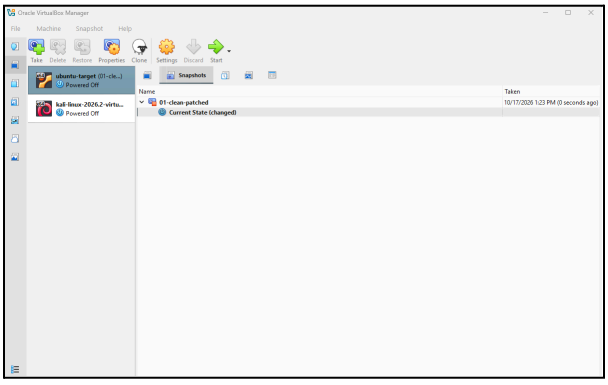
Summary

I built a fully isolated two-VM lab (Kali attacker, Ubuntu target) with a Zeek network sensor on the target, then executed six MITRE ATT&CK techniques and measured which ones Zeek detected, which it only logged, and which ones it missed entirely. The result is a detection coverage matrix backed by real log evidence, plus a Python parser that automates the analysis. The most useful findings are the gaps: a timing-throttled port scan that evades burst-based detection, and an ARP-based host sweep invisible to connection logs.

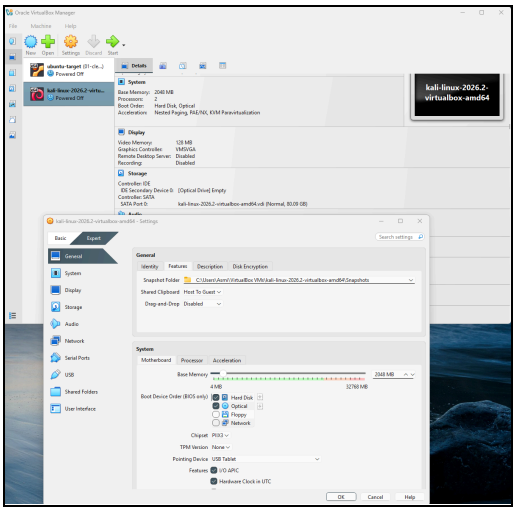
Environment

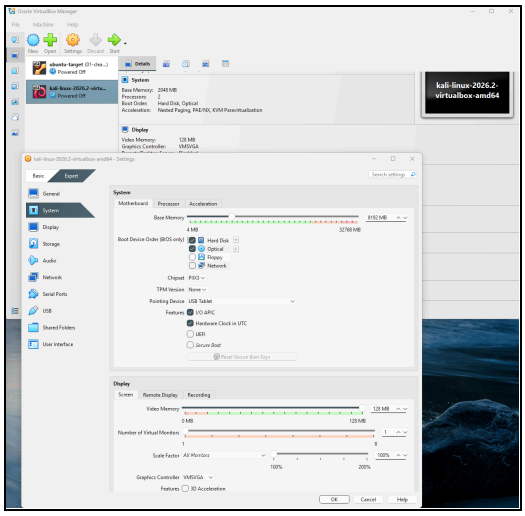
VirtualBox 7.2 | Kali Linux 2026.2 | Ubuntu Server 24.04 LTS | Zeek 9.0

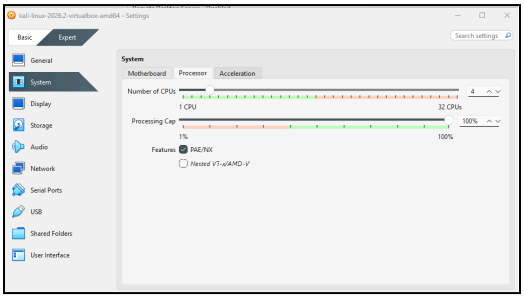


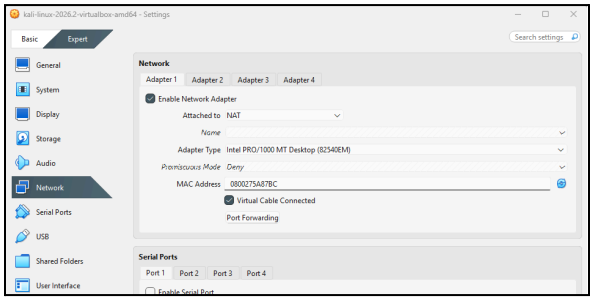


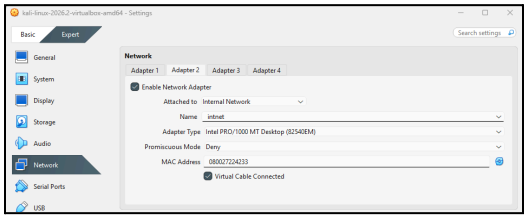
Configure the Kali VM: host to guest so that I can paste commands from my browser into Kali, but nothing copied inside Kali flows back to Windows. It's a one-way valve.

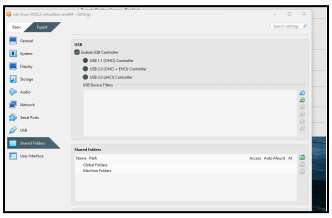

kali-linux-2026.2-virtualbox-amd64

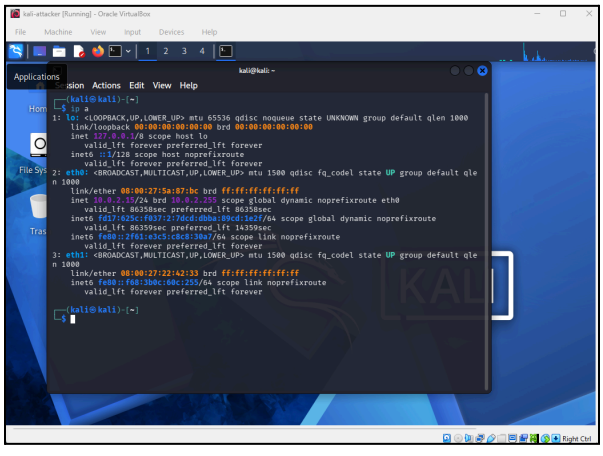

kali-linux-2026.2-virtualbox-amd64

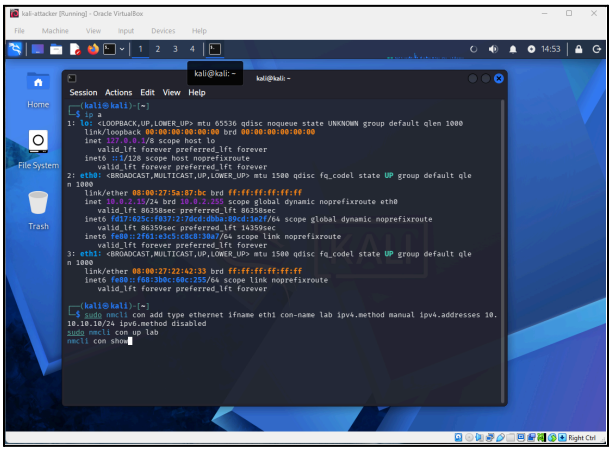


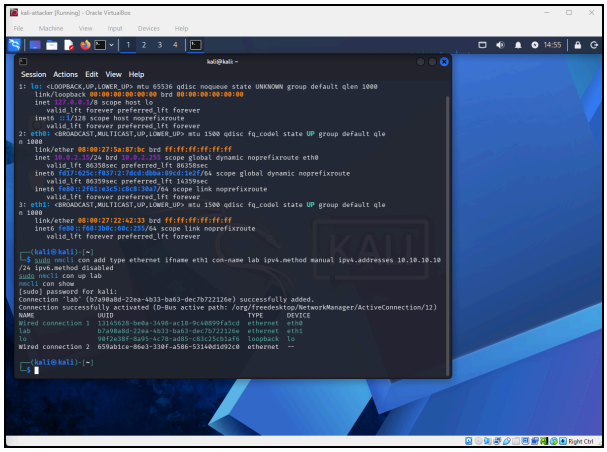


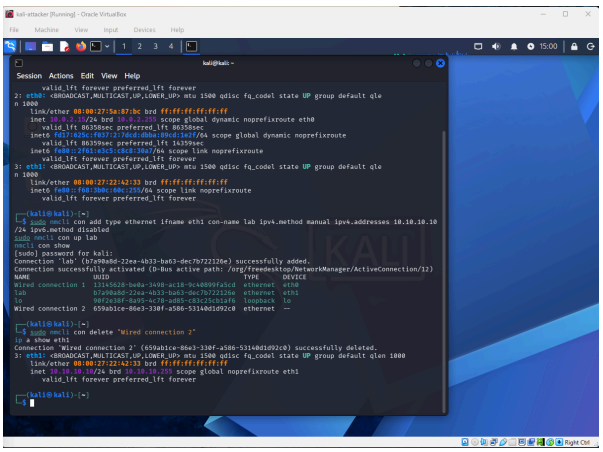




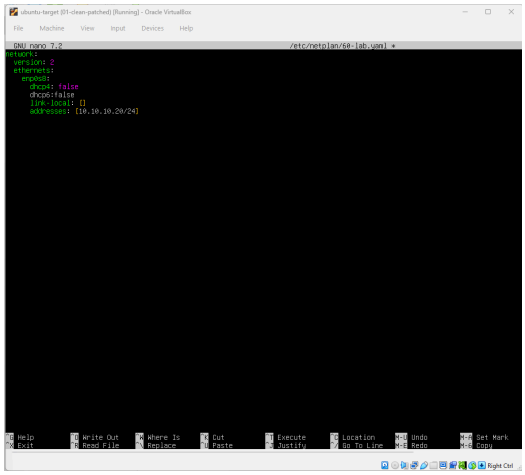




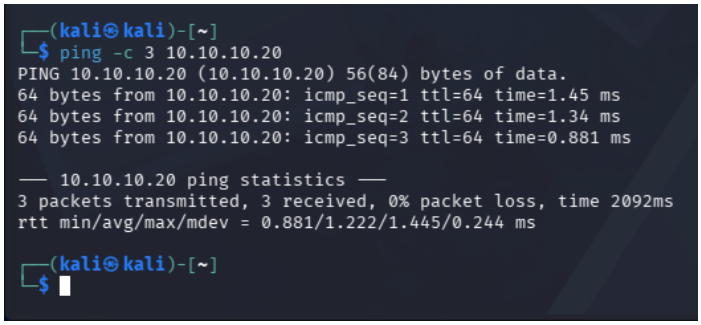


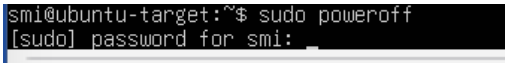


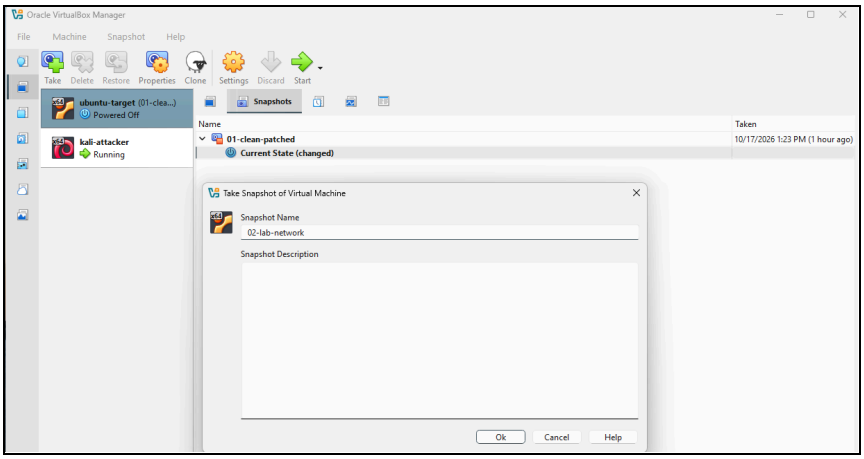
Removed Wired connection 2 so that it doesn’t grab eth1 before lab does, and my lab IP doesn’t quietly vanish

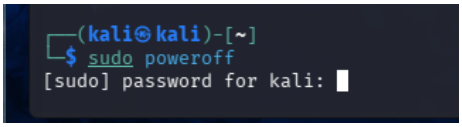


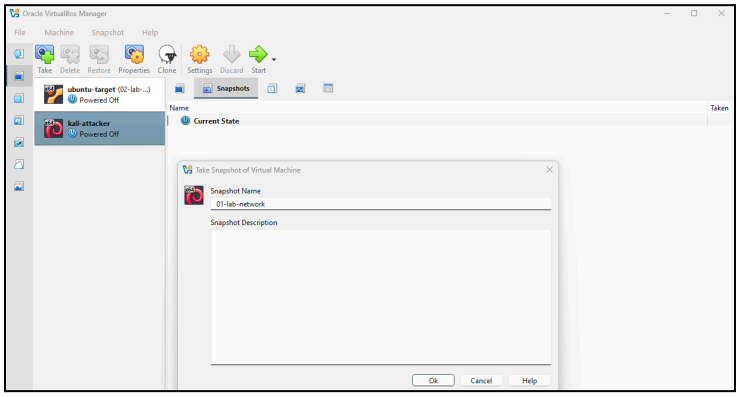




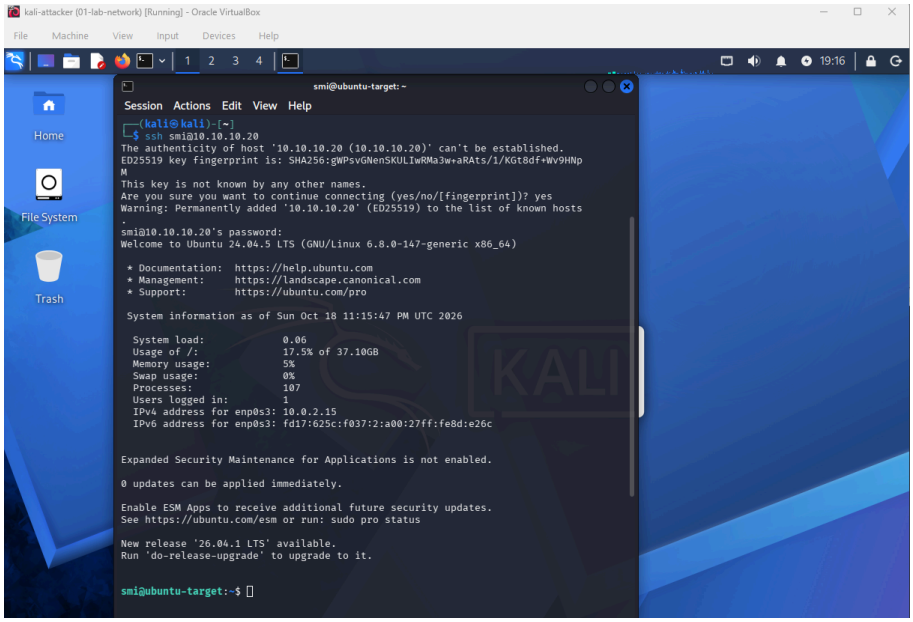






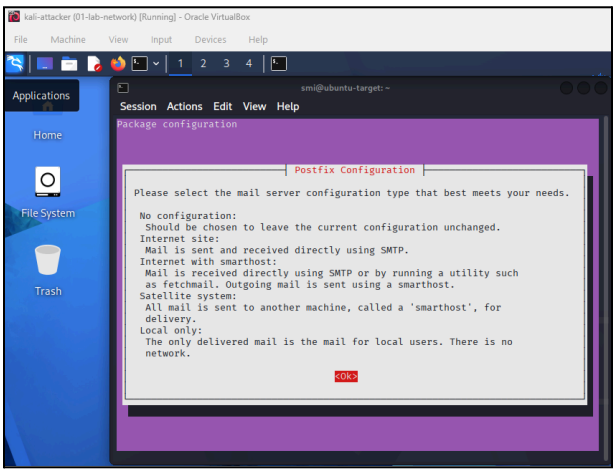


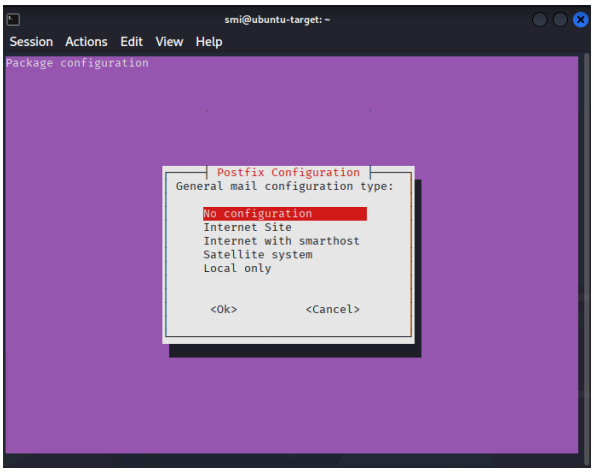
Phase 3, Part 2

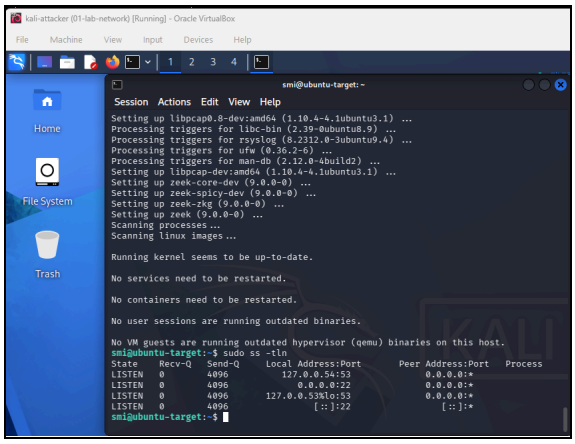


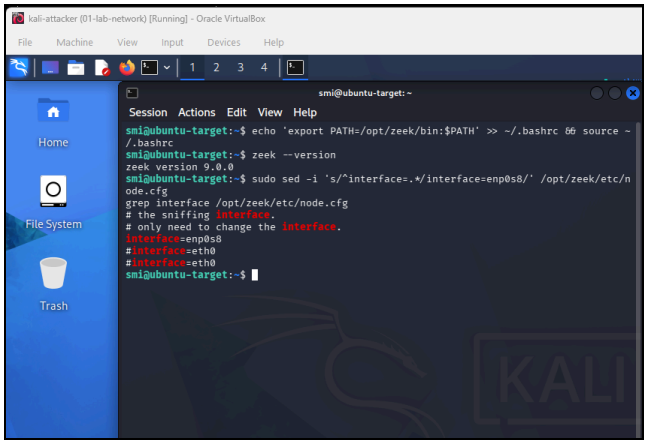
SSH in from kali-attacker to smi@ubuntu-target to make copy pasting commands easier

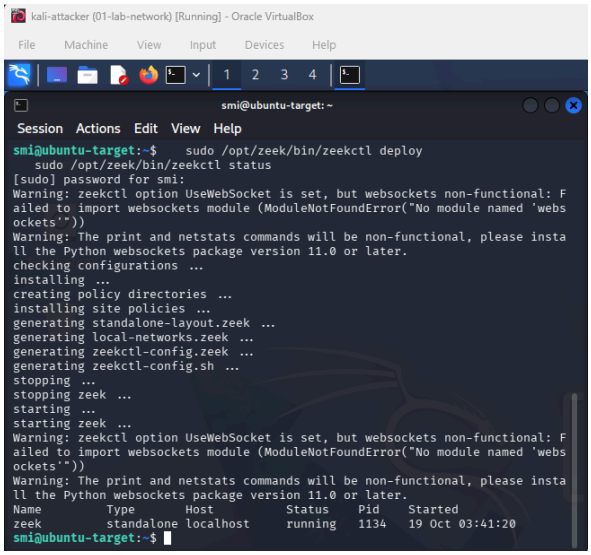
```
smi@ubuntu-target:~$ echo 'deb http://download.opensuse.org/repositories/security:/zeek/xUbuntu_24.04/ /' | sudo tee /etc/apt/sources.list.d/security:zeek.list
curl -fsSL https://download.opensuse.org/repositories/security:zeek/xUbuntu_24.04/Release.key | gpg --dearmor | sudo tee /etc/apt/trusted.gpg.d/security_zeek.gpg > /dev/null
sudo apt update
sudo apt install -y zeek
[sudo] password for smi:
```

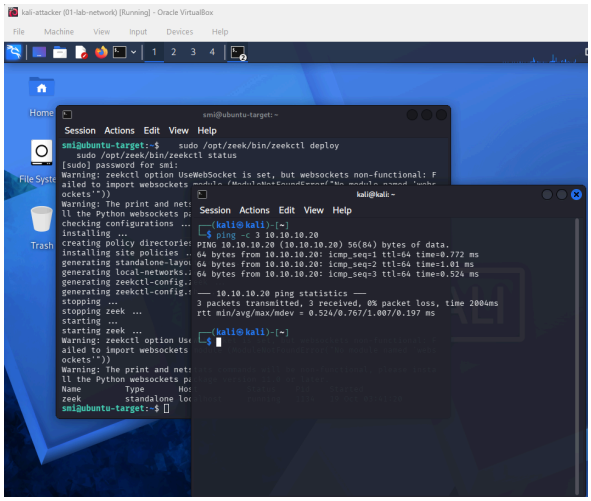


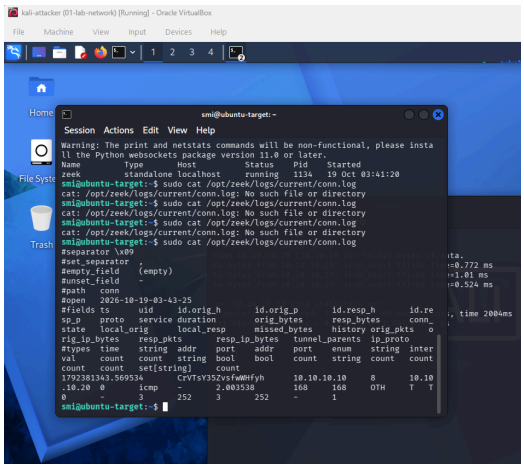


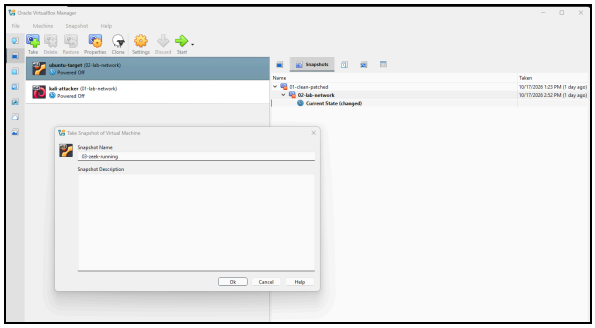

kali-attacker (01-lab-network) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
smi@ubuntu-target: ~
Session Actions Edit View Help
Setting up libpcap0.8-dev:amd64 (1.10.4-4.1ubuntu3.1) ...
Processing triggers for libc-bin (2.39-0ubuntu8.9) ...
Processing triggers for rsyslog (8.2312.0-3ubuntu9.4) ...
Processing triggers for ufw (0.36.2-6) ...
Processing triggers for man-db (2.12.0-4build2) ...
Setting up libpcap-dev:amd64 (1.10.4-4.1ubuntu3.1) ...
Setting up zeek-core-dev (9.0.0-0) ...
Setting up zeek-spicy-dev (9.0.0-0) ...
Setting up zeek-zkg (9.0.0-0) ...
Setting up zeek (9.0.0-0) ...
Scanning processes...
Scanning linux images...
Running kernel seems to be up-to-date.
No services need to be restarted.
No containers need to be restarted.
No user sessions are running outdated binaries.
No VM guests are running outdated hypervisor (qemu) binaries on this host.
smi@ubuntu-target:~$ sudo ss -tln
State Recv-Q Send-Q Local Address:Port Peer Address:Port Process
LISTEN 0 4096 127.0.0.54:53 0.0.0.0:*
LISTEN 0 4096 0.0.0.0:22 0.0.0.0:*
LISTEN 0 4096 127.0.0.53%lo:53 0.0.0.0:*
LISTEN 0 4096 [::]:22 [::]:*
smi@ubuntu-target:~$
Home
File System
Trash

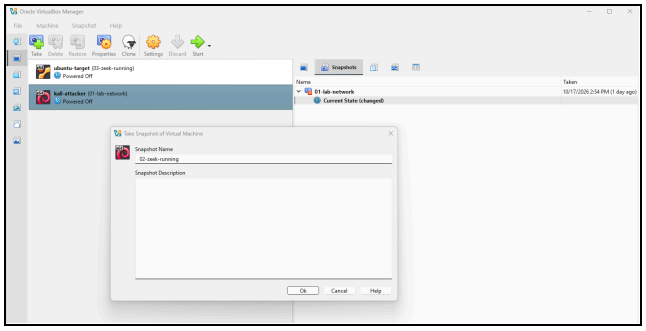

kali-attacker (01-lab-network) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
smi@ubuntu-target: ~
Session Actions Edit View Help
smi@ubuntu-target:~$ echo 'export PATH=/opt/zeek/bin:$PATH' >> ~/.bashrc && source ~/.bashrc
smi@ubuntu-target:~$ zeek --version
zeek version 9.0.0
smi@ubuntu-target:~$ sudo sed -i 's/^interface=.*/interface=enp0s8/' /opt/zeek/etc/node.cfg
grep interface /opt/zeek/etc/node.cfg
the sniffing interface.
only need to change the interface.
interface=enp0s8
#interface=eth0
#interface=eth0
smi@ubuntu-target:~$
Home
File System
Trash


kali-attacker (01-lab-network) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
smi@ubuntu-target: ~
Session Actions Edit View Help
smi@ubuntu-target:~$ sudo /opt/zeek/bin/zeekctl deploy
sudo /opt/zeek/bin/zeekctl status
[sudo] password for smi:
Warning: zeekctl option UseWebSocket is set, but websockets non-functional: Failed to import websockets module (ModuleNotFoundError("No module named 'websockets'"))
Warning: The print and netstats commands will be non-functional, please install the Python websockets package version 11.0 or later.
checking configurations ...
installing ...
creating policy directories ...
installing site policies ...
generating standalone-layout.zeek ...
generating local-networks.zeek ...
generating zeekctl-config.zeek ...
generating zeekctl-config.sh ...
stopping ...
stopping zeek ...
starting ...
starting zeek ...
Warning: zeekctl option UseWebSocket is set, but websockets non-functional: Failed to import websockets module (ModuleNotFoundError("No module named 'websockets'"))
Warning: The print and netstats commands will be non-functional, please install the Python websockets package version 11.0 or later.
Name Type Host Status Pid Started
zeek standalone localhost running 1134 19 Oct 03:41:20
smi@ubuntu-target:~$

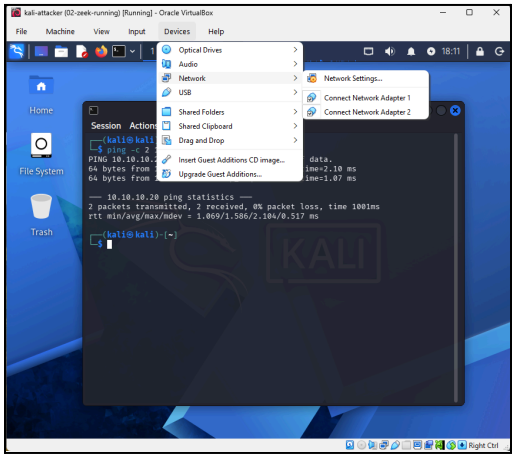

kali-attacker (01-lab-network) [Running] - Oracle VirtualBox
smi@ubuntu-target: ~
kali@kali: ~
Session Actions Edit View Help
sudo /opt/zeek/bin/zeekctl deploy
sudo /opt/zeek/bin/zeekctl status
ping -c 3 10.10.10.20
PING 10.10.10.20 (10.10.10.20) 56(84) bytes of data.
64 bytes from 10.10.10.20: icmp_seq=1 ttl=64 time=0.772 ms
64 bytes from 10.10.10.20: icmp_seq=2 ttl=64 time=1.01 ms
64 bytes from 10.10.10.20: icmp_seq=3 ttl=64 time=0.524 ms
10.10.10.20 ping statistics
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 0.524/0.767/1.007/0.197 ms

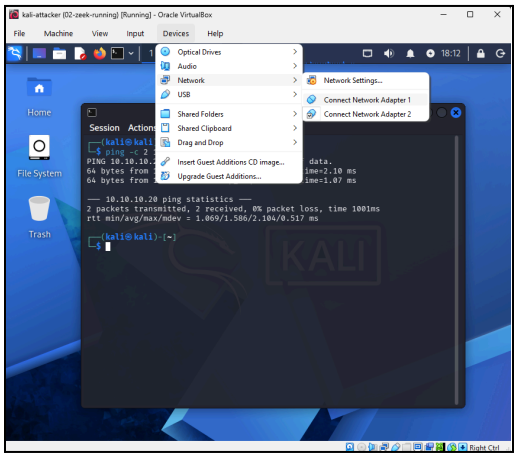

kali-attacker (01-lab-network) [Running] - Oracle VirtualBox
smi@ubuntu-target: ~
Session Actions Edit View Help
Warning: The print and netstats commands will be non-functional, please install the Python websockets package version 11.0 or later.
Name Type Host Status Pid Started
zeek standalone localhost running 1134 19 Oct 03:41:20
smi@ubuntu-target:~$ sudo cat /opt/zeek/logs/current/conn.log
cat: /opt/zeek/logs/current/conn.log: No such file or directory
smi@ubuntu-target:~$ sudo cat /opt/zeek/logs/current/conn.log
cat: /opt/zeek/logs/current/conn.log: No such file or directory
smi@ubuntu-target:~$ sudo cat /opt/zeek/logs/current/conn.log
cat: /opt/zeek/logs/current/conn.log: No such file or directory
smi@ubuntu-target:~$ sudo cat /opt/zeek/logs/current/conn.log
#separator \x09
#set_separator ,
#empty_field (empty)
#unset_field -
#path conn
#open 2026-10-19-03-43-25
#fields ts uid id.orig_h id.orig_p id.resp_h id.resp_p proto service duration orig_bytes resp_bytes conn_state local_orig local_resp missed_bytes history orig_pkts orig_ip_bytes resp_pkts resp_ip_bytes tunnel_parents ip_proto
#types time string addr port addr port enum string interval count count string bool bool count string count count count count set[string] count
1792381343.569534 CrVTsY352vsfWWHfyh 10.10.10.10 8 10.10.10.20 0 icmp - 2.003538 168 168 OTH T T 0 2 252 3 252 - 1
smi@ubuntu-target:~$



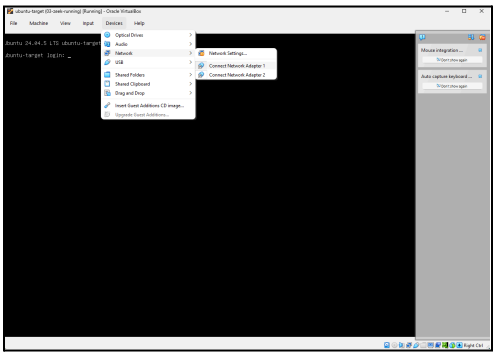


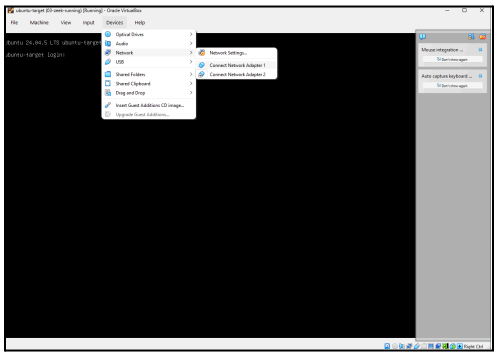
Phase 4: Cut NAT and prove isolation

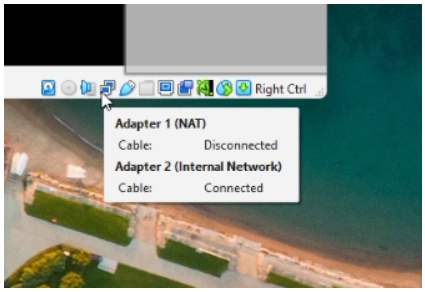




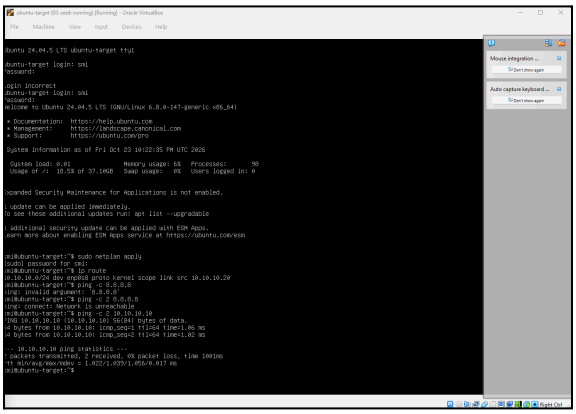
Compare the two icons: Adapter 1 no longer has the checkmark/highlight that Adapter 2 still has. So Kali's NAT is now disconnected and its lab cable is still plugged in.

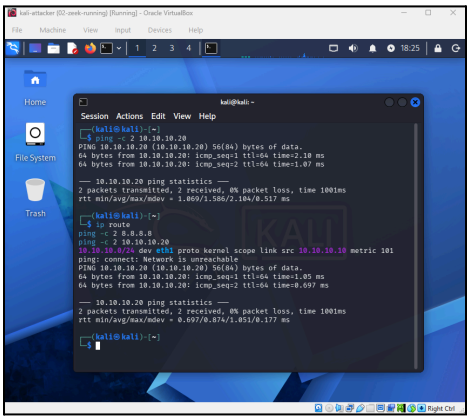


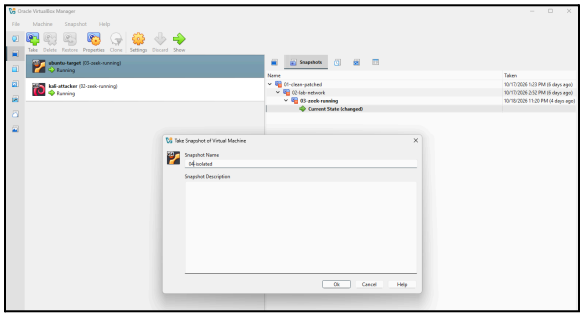


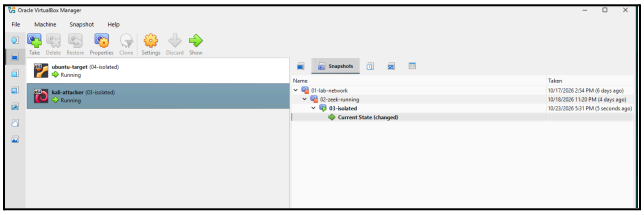


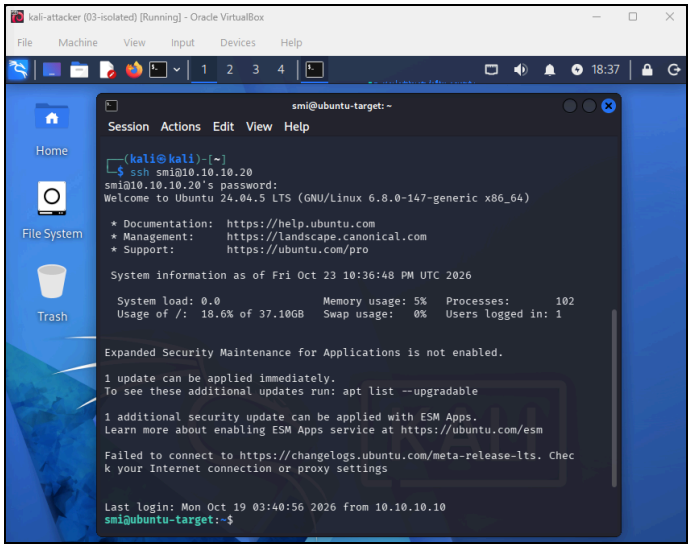
Run the isolation checks

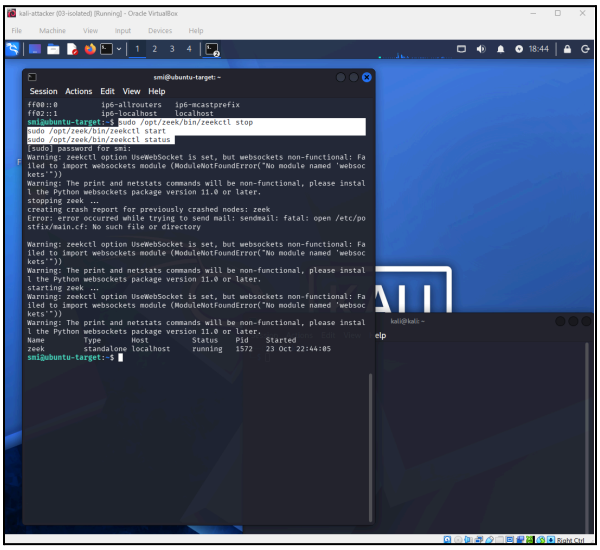


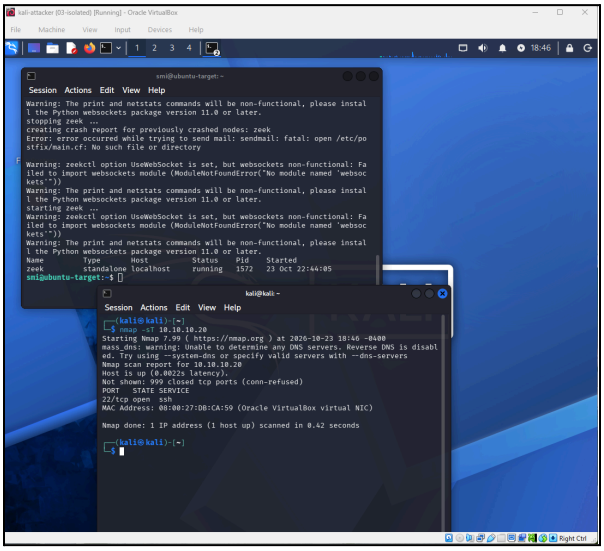




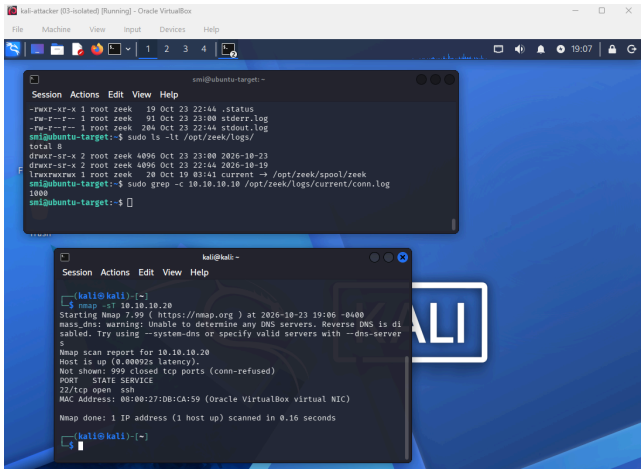



kali-attacker (03-isolated) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
18:37
Home
File System
Trash
smi@ubuntu-target: ~
Session Actions Edit View Help
(kali㉿kali)-[~]
$ ssh smi@10.10.10.20
smi@10.10.10.20's password:
Welcome to Ubuntu 24.04.5 LTS (GNU/Linux 6.8.0-147-generic x86_64)
* Documentation: https://help.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro
System information as of Fri Oct 23 10:36:48 PM UTC 2026
System load: 0.0 Memory usage: 5% Processes: 102
Usage of /: 18.6% of 37.10GB Swap usage: 0% Users logged in: 1
Expanded Security Maintenance for Applications is not enabled.
1 update can be applied immediately.
To see these additional updates run: apt list --upgradable
1 additional security update can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Chec
k your Internet connection or proxy settings
Last login: Mon Oct 19 03:40:56 2026 from 10.10.10.10
smi@ubuntu-target:~$


18:44
smi@ubuntu-target: ~
Session Actions Edit View Help
sudo /opt/zeek/bin/zeekctl stop
sudo /opt/zeek/bin/zeekctl start
sudo /opt/zeek/bin/zeekctl status
[sudo] password for smi:
Warning: zeekctl option UseWebSocket is set, but websockets non-functional: Fa
iled to import websockets module (ModuleNotFoundError("No module named 'websoc
kets'"))
Warning: The print and netstats commands will be non-functional, please instal
l the Python websockets package version 11.0 or later.
stopping zeek ...
creating crash report for previously crashed nodes: zeek
Error: error occurred while trying to send mail: sendmail: fatal: open /etc/po
stfix/main.cf: No such file or directory
starting zeek ...
Name Type Host Status Pid Started
zeek standalone localhost running 1572 23 Oct 22:44:05
smi@ubuntu-target:~$


18:46
smi@ubuntu-target: ~
Session Actions Edit View Help
Warning: The print and netstats commands will be non-functional, please instal
l the Python websockets package version 11.0 or later.
stopping zeek ...
creating crash report for previously crashed nodes: zeek
Error: error occurred while trying to send mail: sendmail: fatal: open /etc/po
stfix/main.cf: No such file or directory
starting zeek ...
Name Type Host Status Pid Started
zeek standalone localhost running 1572 23 Oct 22:44:05
smi@ubuntu-target:~$
kali@kali: ~
Session Actions Edit View Help
(kali㉿kali)-[~]
$ nmap -sT 10.10.10.20
Starting Nmap 7.99 (https://nmap.org) at 2026-10-23 18:46 -0400
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabl
ed. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 10.10.10.20
Host is up (0.0022s latency).
Not shown: 999 closed tcp ports (conn-refused)
PORT STATE SERVICE
22/tcp open ssh
MAC Address: 08:00:27:DB:CA:59 (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.42 seconds
(kali㉿kali)-[~]
$

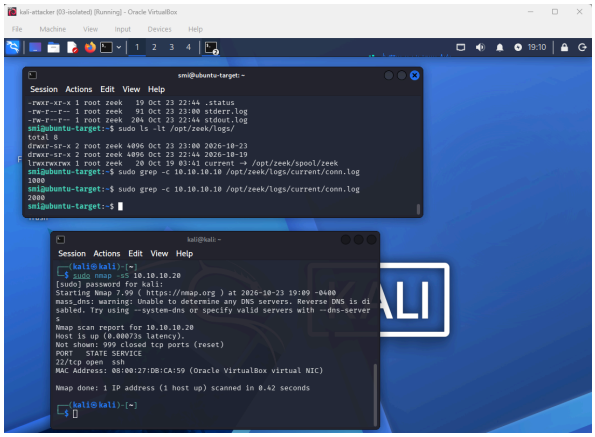
6:08 pm local



Zeek logged a connection for **every one of the 1000 ports** nmap touched, including all 999 closed ones. T1046 via `-sT` → fully detected, 1000 conn.log entries from a single source in under a second. The one-source-to-many-ports burst is an unmistakable scan signature.

T1046 -sT, 19:06 local (23:06 UTC), conn.log, DETECTED (1000 entries).

6:11 pm local

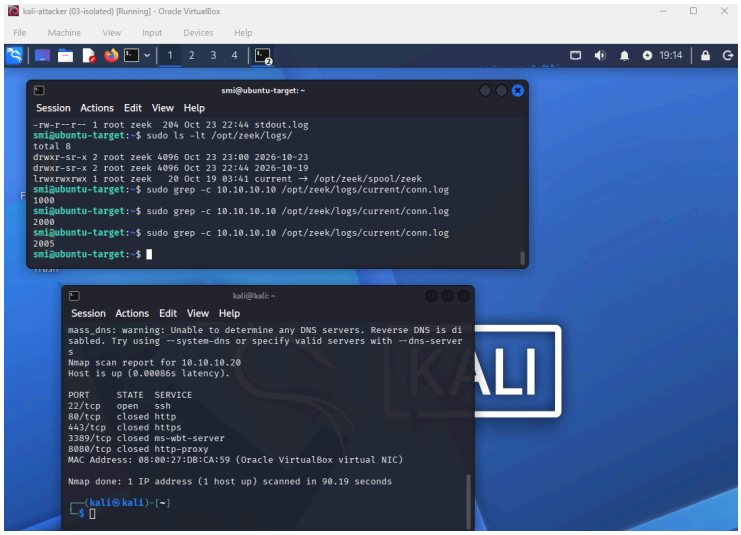


T1046 -sS, 19:09 local (23:09 UTC), conn.log, DETECTED (+1000, total 2000).

6:14pm local

Next is a slow scan. To keep it from taking 15+ minutes, I scan just a few ports:

This scan is about timing. -T1 (“sneaky”) deliberately spaces packets seconds apart to stay under rate-based alarms. Zeek will still log each connection, so it won’t vanish from conn.log. The real question is whether it would slip past detection logic that looks for a burst, since the connections are smeared across minutes instead of arriving all at once.

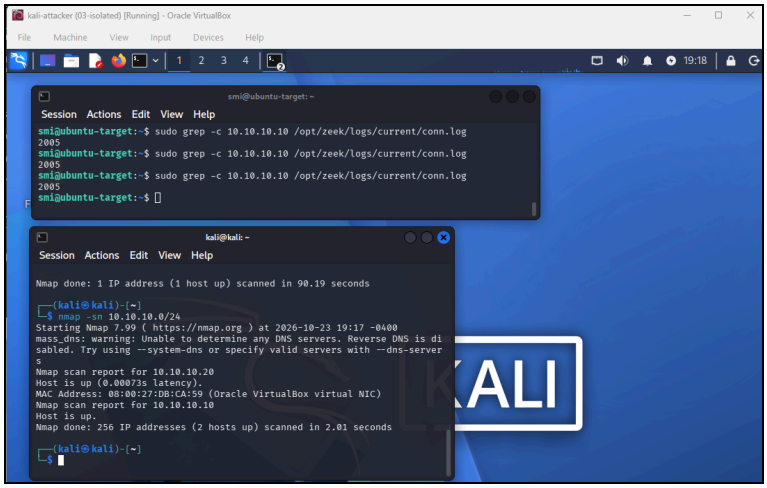


Exactly 5 more, so one per port. Also to note is the timing: nmap says 90.19 seconds to scan just 5 ports. Compared to full 1000-port scans that finished in under half a second each, that’s the -T1 throttle at work: roughly 18 seconds between ports.

Zeek logged every single connection in all three cases, so technically nothing evaded. But detection isn’t just logging. A rule that flags “many connections from one source in a short window” would note the first two but stay silent on the third, because 5 connections spread over 90 seconds looks like normal activity. The slow scan evades detection logic, not logging.

T1046 -T1 slow, ~19:12 local, conn.log, LOGGED but would EVADE burst-detection (5 entries over 90s)

6:18 pm local

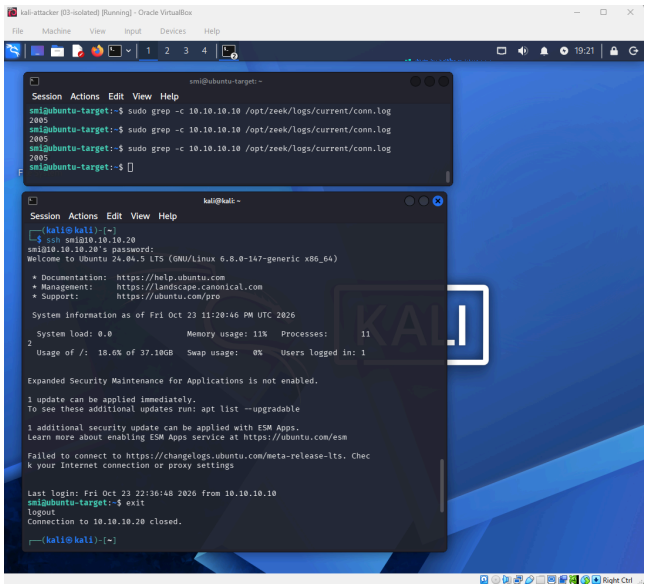


the host sweep produced zero new entries in conn.log

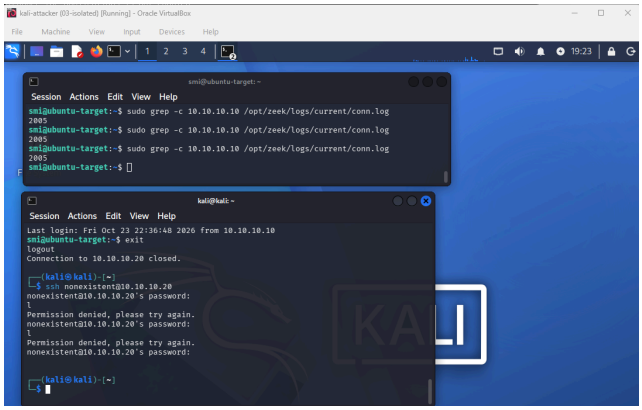
T1018 host sweep, 19:17 local (23:17 UTC), conn.log, MISSED (ARP-based, below TCP layer)

6:22pm local

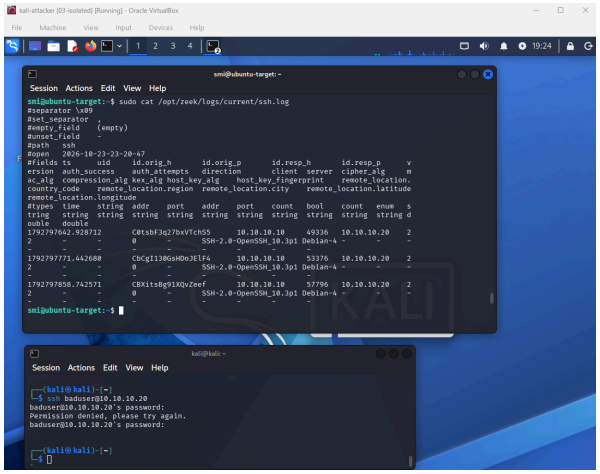
1. Successful login



2. Failed login, wrong username (6:24pm local)

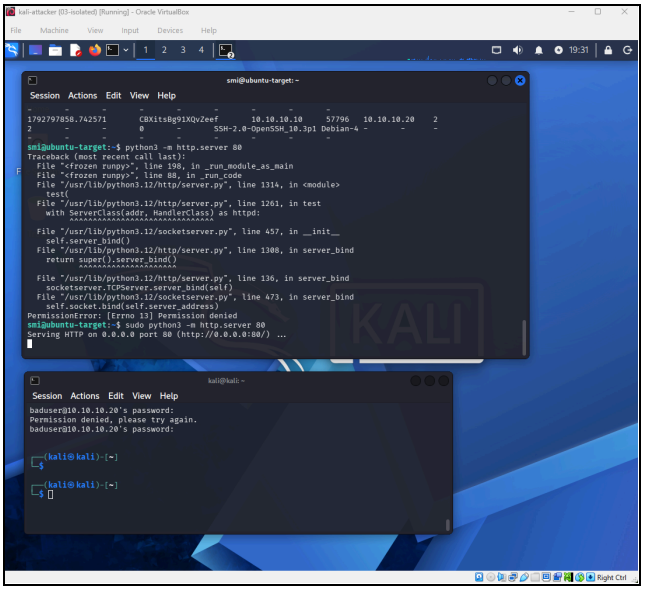


3. Failed login, wrong password (6:25 pm)

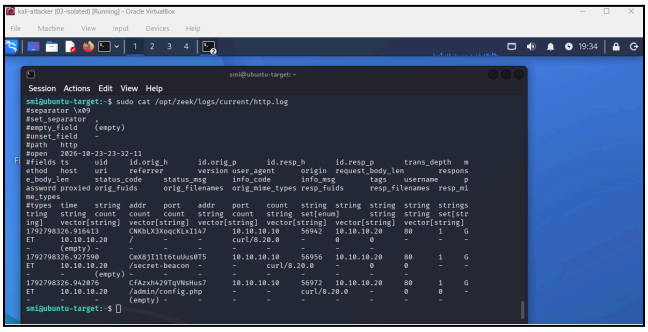


T1021.004 SSH, ~19:24 local (23:24 UTC), ssh.log, DETECTED (3 attempts logged, auth_success undetermined by design)

6:33 pm local





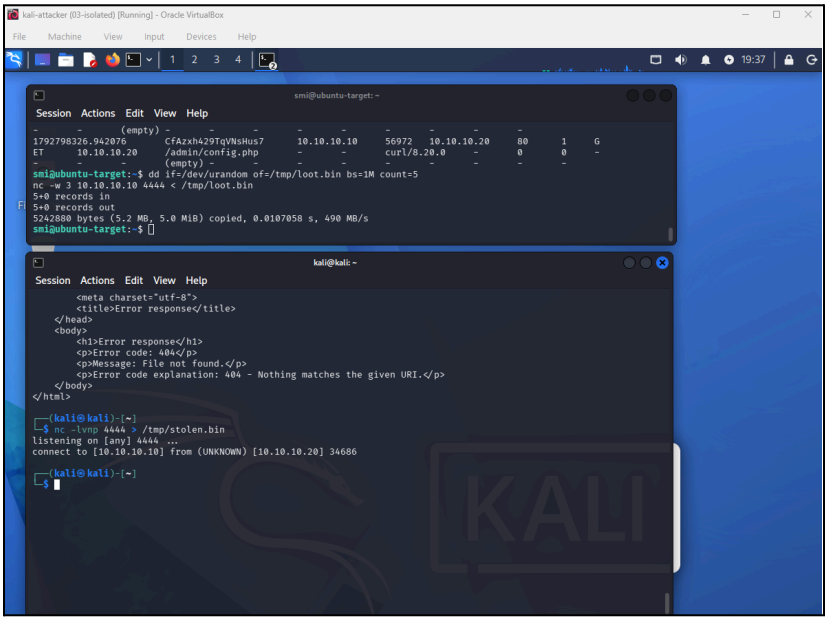


T1071.001 HTTP, 19:32 local (23:33 UTC), http.log, DETECTED (full URIs + curl user-agent captured)

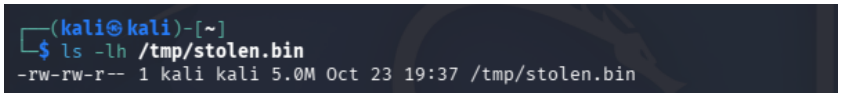
6:38pm local

1. **In the Kali tab, we start a listener (the attacker catching the data) and**

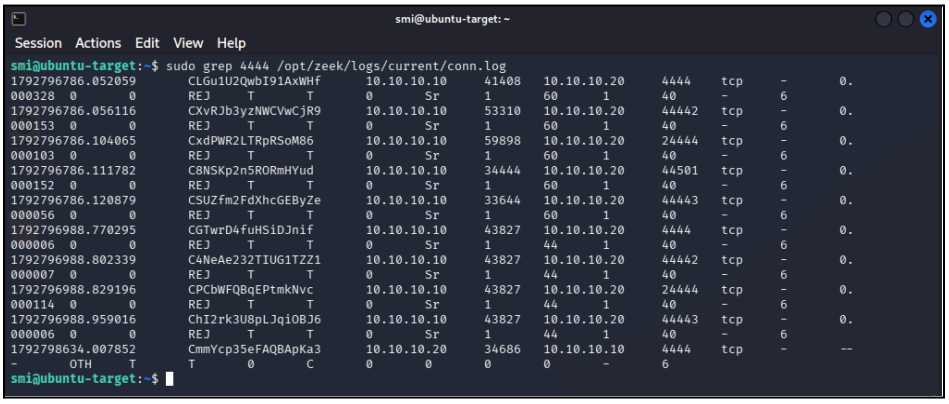
2. **In the SSH tab (on the target), make a 5 MB file and send it:**

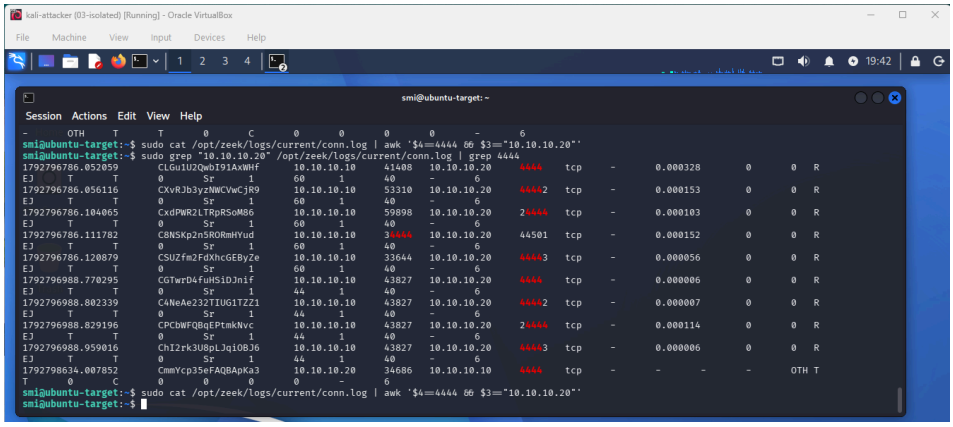


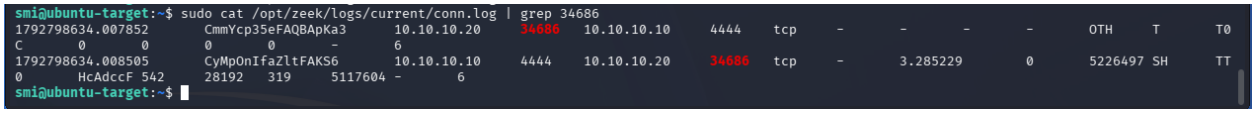
3. Confirm it arrived. On Kali, the listener exits on its own; check the size:



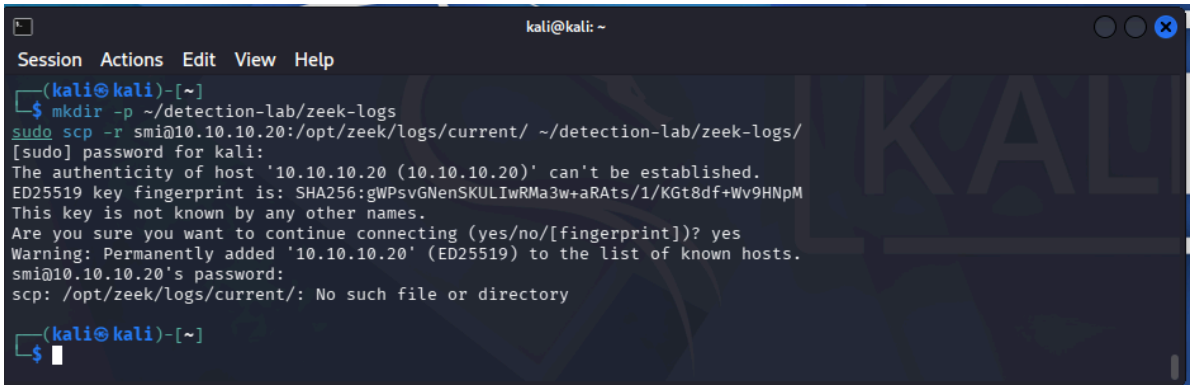
4. **Check Zeek (SSH tab):**

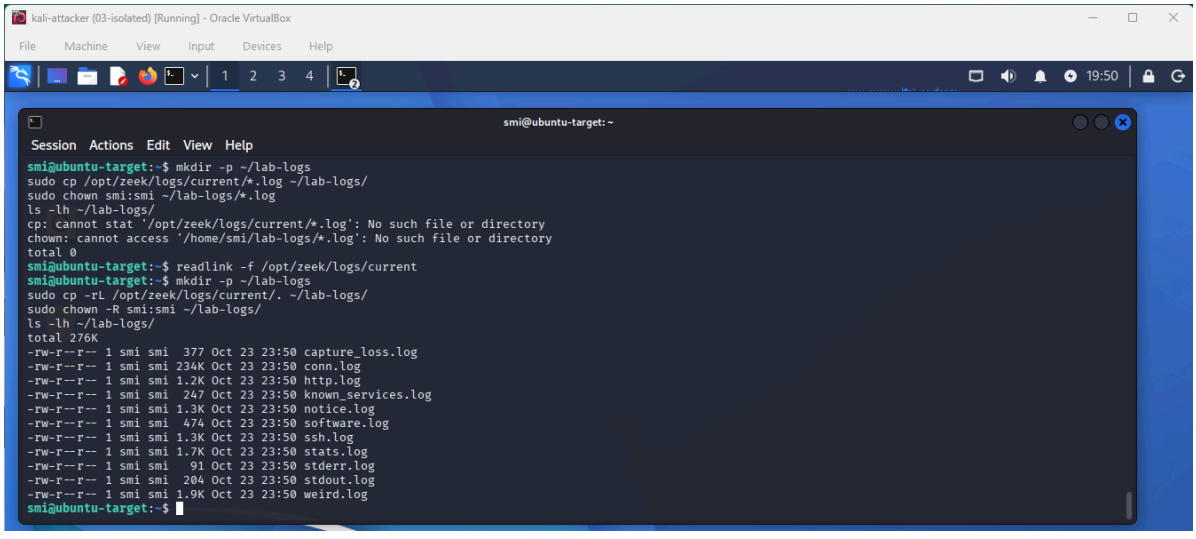


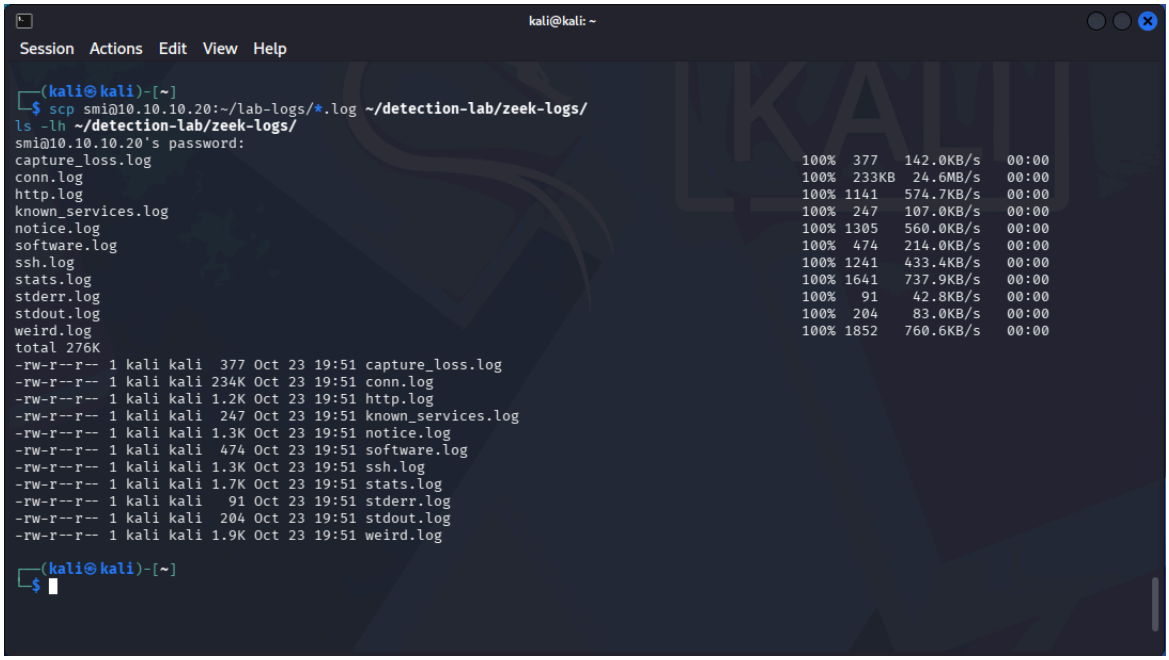




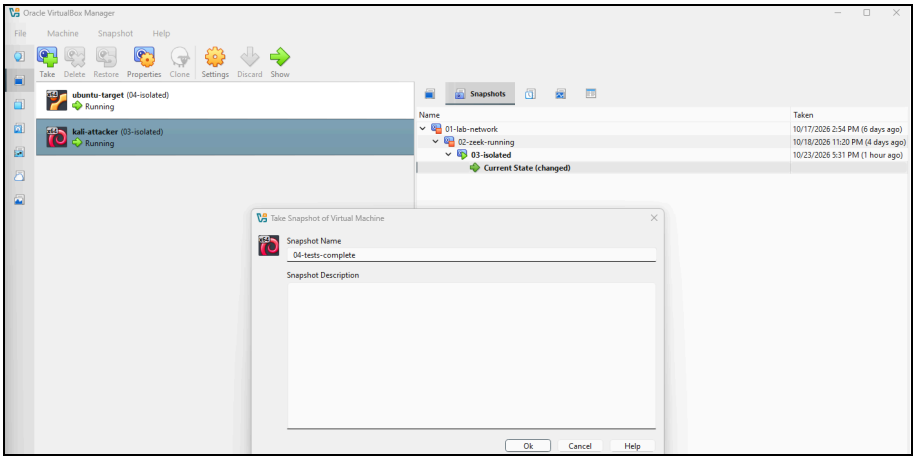
Collect logs

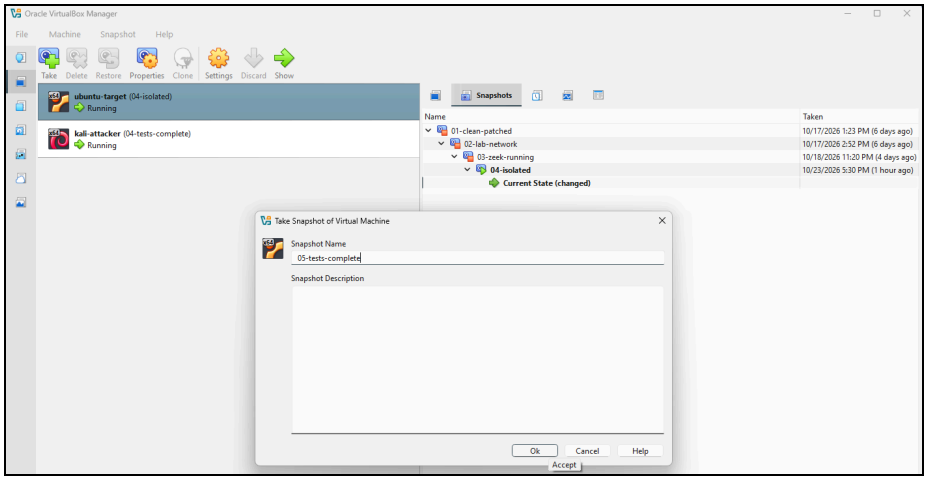




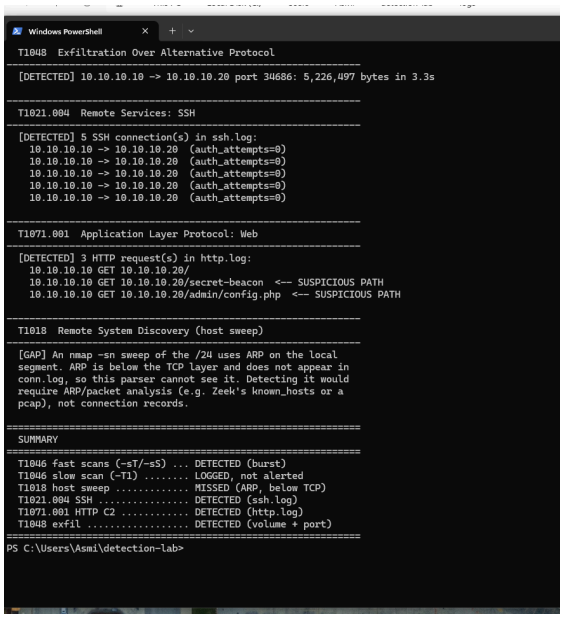


Result: all 6 techniques successfully run, logs collected to ~/detection-lab/zeek-logs and ~/lab-logs.




Oracle VirtualBox Manager
File
Machine
Snapshot
Help
Take
Delete
Restore
Properties
Clone
Settings
Discard
Show
ubuntu-target (04-isolated)
Running
kali-attacker (04-tests-complete)
Running
Snapshots
Name
Taken
01-clean-patched
10/17/2026 1:23 PM (6 days ago)
02-lab-network
10/17/2026 2:52 PM (6 days ago)
03-zeek-running
10/18/2026 11:20 PM (4 days ago)
04-isolated
10/23/2026 5:30 PM (1 hour ago)
Current State (changed)
Take Snapshot of Virtual Machine
Snapshot Name
05-tests-complete
Snapshot Description
Ok
Cancel
Help
Accept


Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.
PS C:\WINDOWS\system32> cd C:\Users\Asmi\detection-lab
PS C:\Users\Asmi\detection-lab> python parse_zeek.py logs
ZEEK DETECTION ANALYSIS — purple-team lab
conn.log records: 2023
T1046 Network Service Discovery (port scan)
[DETECTED] burst scan from 10.10.10.10: >= 20 ports within 5s
connections per source: 10.10.10.10=2022, 10.10.10.20=1
NOTE: the slow nmap -T1 scan is NOT flagged above.
Its 5 connections are spread over ~90s, below the burst
threshold. Every connection is still in conn.log, but a
rate-based detector does not fire. => logged, not alerted.
T1048 Exfiltration Over Alternative Protocol
[DETECTED] 10.10.10.10 -> 10.10.10.20 port 34686: 5,226,497 bytes in 3.3s
T1021.004 Remote Services: SSH
[DETECTED] 5 SSH connection(s) in ssh.log:
10.10.10.10 -> 10.10.10.20 (auth_attempts=0)
10.10.10.10 -> 10.10.10.20 (auth_attempts=0)
10.10.10.10 -> 10.10.10.20 (auth_attempts=0)
10.10.10.10 -> 10.10.10.20 (auth_attempts=0)
10.10.10.10 -> 10.10.10.20 (auth_attempts=0)
T1071.001 Application Layer Protocol: Web
[DETECTED] 3 HTTP request(s) in http.log:
10.10.10.10 GET 10.10.10.20/
10.10.10.10 GET 10.10.10.20/secret-beacon <-- SUSPICIOUS PATH
10.10.10.10 GET 10.10.10.20/admin/config.php <-- SUSPICIOUS PATH
T1018 Remote System Discovery (host sweep)


Windows PowerShell
T1048 Exfiltration Over Alternative Protocol
[DETECTED] 10.10.10.10 -> 10.10.10.20 port 34686: 5,226,497 bytes in 3.3s
T1021.004 Remote Services: SSH
[DETECTED] 5 SSH connection(s) in ssh.log:
10.10.10.10 -> 10.10.10.20 (auth_attempts=0)
10.10.10.10 -> 10.10.10.20 (auth_attempts=0)
10.10.10.10 -> 10.10.10.20 (auth_attempts=0)
10.10.10.10 -> 10.10.10.20 (auth_attempts=0)
10.10.10.10 -> 10.10.10.20 (auth_attempts=0)
T1071.001 Application Layer Protocol: Web
[DETECTED] 3 HTTP request(s) in http.log:
10.10.10.10 GET 10.10.10.20/
10.10.10.10 GET 10.10.10.20/secret-beacon <-- SUSPICIOUS PATH
10.10.10.10 GET 10.10.10.20/admin/config.php <-- SUSPICIOUS PATH
T1018 Remote System Discovery (host sweep)
[GAP] An nmap -sn sweep of the /24 uses ARP on the local
segment. ARP is below the TCP layer and does not appear in
conn.log, so this parser cannot see it. Detecting it would
require ARP/packet analysis (e.g. Zeek's known_hosts or a
pcap), not connection records.
SUMMARY
T1046 fast scans (-sT/-sS) ... DETECTED (burst)
T1046 slow scan (-T1) ..... LOGGED, not alerted
T1018 host sweep ..... MISSED (ARP, below TCP)
T1021.004 SSH ..... DETECTED (ssh.log)
T1071.001 HTTP C2 ..... DETECTED (http.log)
T1048 exfil ..... DETECTED (volume + port)
PS C:\Users\Asmi\detection-lab>